

Identifying errors in proofs - Set 2

TMUA practice questions by topic. Specification Err1, Err2. 7 questions.

Each question has one correct answer. No calculator. Answers and worked solutions are in the separate solutions PDF.

- 1 A student writes the following argument.

I $0.1 > 0.01$.

II Taking logarithms to base 10 of both sides: $-1 > -2$.

III Multiplying both sides by $\log_{10} \frac{1}{2}$, which is negative, reverses the inequality:
 $-\log_{10} \frac{1}{2} < -2\log_{10} \frac{1}{2}$.

Which of the following is correct?

- A** All three lines are correct.
B Line II is wrong: taking logarithms reverses an inequality.
C Line III is wrong: $\log_{10} \frac{1}{2}$ is positive.
D Line II is wrong: $\log_{10} 0.1 = 1$.
E Line I is wrong.
- 2 Claim: if n is an integer and n^2 is divisible by 4, then n is divisible by 4.

I Suppose n^2 is divisible by 4.

II Then $n^2 = 4k$ for some integer k .

III So $n = 2\sqrt{k}$, and since n is an integer $\sqrt{k}$ is an integer.

IV Hence n is even, so $n = 2m$, and $n^2 = 4m^2$ is divisible by 4, so n is divisible by 4.

Which of the following is true?

- A** The claim is false ($n = 2$ is a counterexample) and the first invalid step is on line IV.
B The claim and proof are correct.
C The claim is true but the first error is on line III.
D The claim is false and the first error is on line II.
E The claim is false and the first error is on line I.

3 Claim: $\int_{-1}^1 \frac{1}{x^2} dx = -2$.

I An antiderivative of x^{-2} is $-x^{-1}$.

II So the integral equals $[-\frac{1}{x}]_{-1}^1 = -1 - 1 = -2$.

III Hence the value is -2 .

Which of the following is correct?

A The first error is on line II: the integrand is undefined at $x = 0$, which lies inside the interval, so the fundamental theorem cannot be applied.

B The proof is completely correct.

C The first error is on line I.

D The first error is on line II: $-1 - 1 = 0$.

E The first error is on line III: the value should be $+2$.

4 Claim: for all positive real x , $x + \frac{1}{x} \geq 2$.

I Suppose $x + \frac{1}{x} \geq 2$.

II Multiply by $x > 0$: $x^2 + 1 \geq 2x$.

III So $x^2 - 2x + 1 \geq 0$, i.e. $(x - 1)^2 \geq 0$, which is true.

IV Hence the claim is proved.

Which of the following best describes the argument?

A The argument assumes what it is trying to prove (line I); it would be valid if the steps were reversed, starting from $(x - 1)^2 \geq 0$.

B The argument is completely correct as written.

C The first error is on line II: multiplying by x reverses the inequality.

D The first error is on line III.

E The claim is false.

- 5 Claim: the sum of the first n odd numbers is n^2 .
- I The first odd number is $1 = 1^2$, so the claim holds for $n = 1$.
 - II Suppose the claim holds for some n , i.e. $1 + 3 + \cdots + (2n - 1) = n^2$.
 - III Adding the next odd number, $2n + 1$, gives $n^2 + 2n + 1 = (n + 1)^2$.
 - IV So the claim holds for $n + 1$, and hence for all positive integers n .

Which of the following is correct?

- A The proof is completely correct.
- B The first error is on line II: you cannot assume the claim.
- C The first error is on line III: the next odd number is $2n - 1$.
- D The first error is on line IV: it only shows the claim for $n + 1$.
- E The first error is on line I.

- 6 Claim: if $f'(a) = 0$ and $f''(a) = 0$ then f has a point of inflexion at $x = a$.

Which of the following shows the claim is false?

- A $f(x) = x^3 - 3x$ at $a = 1$
- B $f(x) = x$ at $a = 0$
- C $f(x) = x^2$ at $a = 0$
- D $f(x) = x^3$ at $a = 0$
- E $f(x) = x^4$ at $a = 0$

- 7 Claim: the function $f(x) = x^3 - 3x^2 + 3x$ is increasing for all real x .

- I $f'(x) = 3x^2 - 6x + 3 = 3(x - 1)^2$.
- II So $f'(x) \geq 0$ for all x , with equality only at $x = 1$.
- III Since $f'(x) > 0$ for all x , f is increasing.

Which of the following is true?

- A Line III overstates what was shown: $f'(1) = 0$; but f is still increasing for all x , so the claim is true and the proof needs a better justification.
- B The proof is completely correct.
- C The first error is on line I.
- D The first error is on line II: $f'(x)$ can be negative.
- E The claim is false: f has a turning point at $x = 1$.